

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Blu-ray
Manufacturer: Arcam
Firmware Version: N/A
Model(s): UDP411

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.2

Version History

Module Version	Date	Notes
1_0_0_0	1/16/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5

Control Commands, States, and Qualifiers:

Keypad	'0' – '9'		
MenuNavigation	'Popup Menu' 'Right' 'Home' 'OK'	'Top Menu' 'Up' 'Cancel'	'Left' 'Down' 'Return'
Power	'On'	'Off'	
Random	None		
RCEmulation	'Standby' '>>' 'Display' 'Trim' 'Mode' 'Setup' 'Yellow/C'	'Search' ' <' 'Status' 'Angle' 'Audio' 'Red/A' 'Blue/D'	'<<' '> ' 'HDMI' 'Zoom' 'Subtitles' 'Green/B' 'PIP Audio'
Repeat	None		
Transport	'Play' 'Previous' 'Fast Forward'	'Stop' 'Next' 'A-B'	'Pause' 'Rewind'
Tray	None		

Status Available:

ConnectionStatus	'Connected'	'Disconnected'	
ElapsedTime	String		
Input	'Blu-ray' 'Data Disc' 'No Media'	'DVD-video' 'USB Media'	'CD' 'Network Media'
PlaybackStatus	'Stop' 'Rewind' 'Other'	'Play' 'Fast Forward'	'Pause' 'Resume-stop'
Power	'On'	'Off'	
RandomStatus	'On'	'Off'	
RepeatStatus	'One'	'All'	
Title_Chapter_TrackStatus¹	0 – 65535		
['Source']	'DVD-Video'	'Blu-ray'	'Other'
['Type']	'Title'	'Chapter'	'Track'
TrayStatus	'Open'	'Closed'	

¹ Values shown for this status are only accurate if the Source value selected is equal to the one presented for Input.

Cable and Adapter Requirements:

Captive Screw to Male DB9 RS-232 Serial Cable

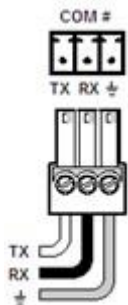
Notes for the Device:

Serial communication:

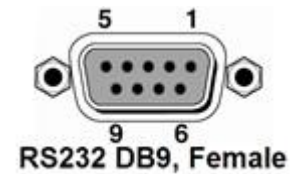
Port Type: RS-232
Baud Rate: 38400
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		3	TxD
RxD		2	RxD
GND		5	GND



General Notes:

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	50000
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Keypad 0	!\x01\x08\x02\x19\x00\x0D
Keypad 1	!\x01\x08\x02\x19\x01\x0D
Keypad 2	!\x01\x08\x02\x19\x02\x0D
Keypad 3	!\x01\x08\x02\x19\x03\x0D
Keypad 4	!\x01\x08\x02\x19\x04\x0D
Keypad 5	!\x01\x08\x02\x19\x05\x0D
Keypad 6	!\x01\x08\x02\x19\x06\x0D
Keypad 7	!\x01\x08\x02\x19\x07\x0D
Keypad 8	!\x01\x08\x02\x19\x08\x0D
Keypad 9	!\x01\x08\x02\x19\x09\x0D
Menu Navigation Cancel	!\x01\x08\x02\x19:\x0D
Menu Navigation Down	!\x01\x08\x02\x19U\x0D
Menu Navigation Home	!\x01\x08\x02\x19\x0B\x0D
Menu Navigation Left	!\x01\x08\x02\x19Q\x0D
Menu Navigation OK	!\x01\x08\x02\x19W\x0D
Menu Navigation Popup Menu	!\x01\x08\x02\x19C\x0D
Menu Navigation Return	!\x01\x08\x02\x19H\x0D
Menu Navigation Right	!\x01\x08\x02\x19P\x0D
Menu Navigation Top Menu	!\x01\x08\x02\x19B\x0D
Menu Navigation Up	!\x01\x08\x02\x19V\x0D
Power Off	!\x01\x08\x02\x19 \x0D
Power On	!\x01\x08\x02\x19{\x0D
Random None	!\x01\x08\x02\x19@\x0D
RC Emulation <	!\x01\x08\x02\x19!\x0D
RC Emulation <<	!\x01\x08\x02\x192\x0D
RC Emulation >	!\x01\x08\x02\x19 \x0D
RC Emulation >>	!\x01\x08\x02\x194\x0D
RC Emulation Angle	!\x01\x08\x02\x19I\x0D
RC Emulation Audio	!\x01\x08\x02\x19E\x0D
RC Emulation Blue/D	!\x01\x08\x02\x19n\x0D
RC Emulation Display	!\x01\x08\x02\x19\x12\x0D
RC Emulation Green/B	!\x01\x08\x02\x19I\x0D
RC Emulation HDMI	!\x01\x08\x02\x19X\x0D
RC Emulation Mode	!\x01\x08\x02\x19M\x0D
RC Emulation PIP Audio	!\x01\x08\x02\x19j\x0D
RC Emulation Red/A	!\x01\x08\x02\x19k\x0D
RC Emulation Search	!\x01\x08\x02\x19L\x0D
RC Emulation Setup	!\x01\x08\x02\x19J\x0D
RC Emulation Standby	!\x01\x08\x02\x19\x0C\x0D
RC Emulation Status	!\x01\x08\x02\x19K\x0D
RC Emulation Subtitles	!\x01\x08\x02\x19A\x0D
RC Emulation Trim	!\x01\x08\x02\x19%\x0D
RC Emulation Yellow/C	!\x01\x08\x02\x19m\x0D
RC Emulation Zoom	!\x01\x08\x02\x19D\x0D
Repeat None	!\x01\x08\x02\x19\x1D\x0D

Transport A-B	!\x01\x08\x02\x19i\x0D
Transport Fast Forward	!\x01\x08\x02\x194\x0D
Transport Next	!\x01\x08\x02\x19 \x0D
Transport Pause	!\x01\x08\x02\x190\x0D
Transport Play	!\x01\x08\x02\x195\x0D
Transport Previous	!\x01\x08\x02\x19!\x0D
Transport Rewind	!\x01\x08\x02\x192\x0D
Transport Stop	!\x01\x08\x02\x196\x0D
Tray None	!\x01\x08\x02\x19-\x0D

Appendix B. Query Commands

Elapsed Time	!\x01(\x01\xF0\x0D
Input	!\x01,\x01\xF0\x0D
Playback Status	!\x01)\x01\xF0\x0D
Power	!\x01\x00\x01\xF0\x0D
Random Status	!\x01)\x01\xF0\x0D
Repeat Status	!\x01)\x01\xF0\x0D
Title/Chapter/Track Status Blu-ray Chapter	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status Blu-ray Title	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status Blu-ray Track	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status DVD-Video Chapter	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status DVD-Video Title	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status DVD-Video Track	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status Other Chapter	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status Other Title	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status Other Track	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status Source Blu-ray	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status Source DVD-Video	!\x01-\x01\xF0\x0D
Title/Chapter/Track Status Source Other	!\x01-\x01\xF0\x0D
Tray Status	!\x01)\x01\xF0\x0D